

SIDDHARTH SINGH BHADAURIA

Senior Cloud & Data Engineer | Generative AI & Cloud Infrastructure Specialist

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Portfolio: siddharthsbhadauria.github.io/portfolio/ • **GitHub:** github.com/siddharthsbhadauria • **LinkedIn:** linkedin.com/in/siddharthsbhadauria

PROFESSIONAL SUMMARY

Senior Cloud and Data Engineer with over 11.5 years of experience architecting enterprise cloud platforms, cloud data pipelines and Generative AI systems across Google Cloud Platform (GCP), Microsoft Azure and AWS. Proven track record deploying GCP multi-cloud unified control plane logging (ATLAS 2.0, 50M+ daily log events), CMEK-encrypted storage, GCS Object Lifecycle Management, BigQuery Linked Datasets and Kafka Connect Databus pipelines, large-scale AWS EKS cluster modules (supporting 3,600+ deployments and 700+ applications) and enterprise Retrieval-Augmented Generation (RAG) platforms serving 75,000+ users. Active open-source author and certified Google Professional ML Engineer & Data Engineer leveraging AI-assisted coding tools (GitHub Copilot, Claude Code and Google Antigravity) to deliver high-performance cloud infrastructure and automated data solutions.

TECHNICAL CORE COMPETENCIES

- **Cloud Infrastructure & DevOps:** Google Cloud Platform (GCP), Microsoft Azure, Amazon Web Services (AWS), AWS EKS, Multi-Cloud Unified Control Plane (ATLAS 2.0), GCP Cloud Asset Inventory (CAI), Log Routers, Log Sinks, Log Buckets, GCS Object Lifecycle Management (Hot to Archive Tiering), CMEK (Customer-Managed Encryption Keys), Cloud KMS, BigQuery Linked Datasets, Terraform, Docker, Kubernetes (AKS, GKE), Azure Container Apps, Google Private Service Connect, Azure API Management, GitHub Actions, GitLab CI/CD
- **Generative AI & LLMs:** Retrieval-Augmented Generation (RAG), Vector Databases (Postgres pgvector, Qdrant, ChromaDB), Confidential AI DeepEval Framework, Prompt Engineering, Azure OpenAI, GCP Vertex AI, Gemini 1.5 Pro/Flash, Llama 3, Mistral 7B
- **AI Assistants & Developer Tools:** GitHub Copilot, Claude Code, Google Antigravity, VS Code, Git, PyTest, Go Test
- **Data Engineering & Streaming:** Data Architecture, Enterprise Observability, Kafka Connect Source Connectors, Enterprise Databus, BigQuery Linked Datasets, DuckDB, Polars, Great Expectations Data Contracts, BigQuery, Pub/Sub, Splunk Log Ingestion, Cloud Functions, Cloud Scheduler, Azure Data Factory (ADF), dbt (Data Build Tool), Apache Kafka, Apache NiFi, AbInitio, IBM DataStage
- **Databases & Storage:** PostgreSQL, Azure SQL Server, Oracle 12c, Teradata, Sybase, Azure Cosmos DB, AWS S3, DynamoDB, RDS, Google Cloud Storage (GCS), Azure Blob Storage
- **Security & Compliance:** CMEK (Customer-Managed Encryption Keys), Cloud KMS, PII Identification & Masking, Fernet Cryptography, GPG Key Management & Rotation, Private DNS Forwarding, Azure Key Vault, Enterprise Data Governance
- **Languages & Scripting:** Python, Go, SQL, Unix Shell Scripting, PySpark, Agile / Scrum Practices

OPEN-SOURCE ENGINEERING PROJECTS & PORTFOLIO

Cloud-FinOps-RAG | siddharthsbhadauria.github.io/cloud-finops-rag/

- Engineered an autonomous multi-cloud AI tokenomics price benchmark evaluator automated via GitHub Actions CRON pipelines, tracking 15+ LLM provider endpoints.
- Built data contract validation using Great Expectations assertions, analytics processing via Polars and DuckDB, vector search indexing in ChromaDB and published automated briefs to GitHub Pages.

Auto-DataPulse | github.com/siddharthsbhadauria/auto-datapulse

- Architected an automated IoT data telemetry and lakehouse pipeline collecting metrics across 20+ containerized sensors into an embedded DuckDB and Parquet storage layer.
- Implemented schema assertion contracts using Great Expectations, data transformation workflows via dbt-core and Docker containerization.

Awesome-AI-Discoveries | github.com/siddharthsbhadauria/awesome-ai-discoveries

- Built an automated AI agent repository discovery pipeline tracking emerging LLM frameworks, RAG systems and developer tools.

Personal Interactive Engineering Portfolio | siddharthsbhadauria.github.io/portfolio/

- Designed and deployed a responsive cloud engineering portfolio showcasing architecture patterns, live project dashboards and open-source contributions.

PROFESSIONAL EXPERIENCE

Senior Cloud Infrastructure & Observability Engineer |
Deloitte (Client: JPMorgan Chase & Co.)

London, UK | June 2026 - Present

JPMorgan Chase - ATLAS 2.0 (GCP Unified Control Plane Observability, Logging & Inventory)

- Architected and implemented the GCP observability, logging and asset inventory module within JPMorgan Chase's ATLAS 2.0 Multi-Cloud Unified Control Plane platform (extending existing AWS and Azure governance layers).

- Configured GCP observability telemetry workflows, establishing Log Buckets, Log Routers and Log Sinks to aggregate and stream over 50M+ daily log events to BigQuery and Pub/Sub with sub-second ingestion latency.
- Configured Kafka Connect Source Connectors to stream log events from GCP Pub/Sub to enterprise Databus streams for downstream distribution.
- Configured Google Cloud Storage Object Lifecycle Management rules on log buckets to automatically transition telemetry objects from hot tier storage to archive tier (Nearline, Coldline and Archive), optimizing long-term retention costs.
- Enforced enterprise banking security and encryption compliance by securing GCP Logging Buckets with Customer-Managed Encryption Keys (CMEK) via Cloud KMS.
- Created BigQuery Linked Datasets directly connected to GCP Logging Buckets, reducing log storage overhead by 40% while enabling real-time SQL analysis without data duplication.
- Built automated asset inventory tracking pipelines utilizing Cloud Asset Inventory (CAI), Cloud Functions and Cloud Scheduler to poll and export resource asset metadata across 10,000+ cloud resources to V12 datastores.
- Engineered log routing feeds to direct enterprise audit telemetry to Splunk for SIEM security monitoring, managing data pipeline interfaces up to Splunk ingestion boundaries.

Parental Leave | Career Break

London, UK | Oct 2025 - May 2026

- Took planned parental leave following the completion of the AWS EKS cluster deployment project.

Senior Consultant & DevOps Engineer | Deloitte (Client: JPMorgan Chase & Co.)

London, UK | Aug 2024 - Sept 2025

JPMorgan Chase (Terraform AWS EKS & Microservices Platform)

- Created, managed and continuously improved a Terraform AWS EKS cluster module alongside a Stack microservices blueprint connecting EKS to AWS resources including S3, DynamoDB, RDS, MSK, SQS and SNS, reducing provisioning time by 65%.
- Scaled the EKS Cluster module to over 3,600+ active deployments supporting 700+ microservices applications across 20+ Lines of Business (LoBs) across the bank.
- Evolved EKS cluster module functionality into a multi-tenant deployment model to streamline resource allocation and tenant isolation.
- Managed multiple Kubernetes version upgrades from v1.19 through v1.34 with documented upgrade paths for end users, supporting 50+ zero-downtime Production releases.
- Built automated testing and acceptance test suites using Python and Go to validate module enhancements deployed on EKS clusters.

Senior Consultant | Deloitte (Internal: PairD GenAI Platform)

London, UK | May 2024 - Aug 2024

Deloitte PairD (Enterprise Generative AI Platform)

- Co-engineered Deloitte's internal Generative AI platform serving 75,000+ employees across the firm, handling over 100,000+ daily queries across OpenAI GPT and Google Gemini models.
- Architected secure networking using Google Private Service Connect (PSC) and custom DNS forwarding to achieve 100% network isolation with zero public internet exposure, ensuring all data flows routed strictly through internal Deloitte infrastructure.
- Designed scalable Retrieval-Augmented Generation (RAG) pipelines, configuring vector databases (Postgres with pgvector, Qdrant) and tuning prompt templates to optimize model outputs.
- Led containerization workstream, converting Azure cloud deployments into modular Infrastructure as Code via Terraform and ARM templates.
- Configured Docker and Kubernetes containerized services deployed to Azure Container Apps and Azure Kubernetes Service (AKS).
- Accelerated frontend and backend engineering workflows by 35% leveraging AI-assisted coding tools including GitHub Copilot, Claude Code and Google Antigravity.
- Built automated DevOps deployment pipelines with GitHub Actions and Azure Container Registries, exposing backend services securely through Azure API Management.

Senior Consultant | Deloitte (Internal: GenAI LLM Evaluation)

London, UK | Apr 2024 - May 2024

GenAI LLM Evaluation Framework

- Built an automated LLM evaluation pipeline using Confident AI DeepEval to quantify model quality across 4 core metrics (Answer Relevancy, Contextual Accuracy, Completeness and Latency).
- Deployed models including Mistral 7B, Llama 3 and Gemini 1.0 Pro on GCP Vertex AI for benchmark testing and evaluation.
- Authored Python automation scripts to harvest and visualize evaluation metrics, reducing model benchmarking cycle time from 2 weeks to under 4 hours.

Architect & Senior Developer | Deloitte (Internal: DataSAFE Accelerator)

London, UK | Dec 2023 - Mar 2024

DataSAFE & M&A Data Transfer Accelerator

- Designed data processing architecture for 10TB+ of GPG-encrypted datasets, incorporating schema validation, file integrity checks and automated decryption with 100% data integrity.
- Engineered Python data protection scripts to detect and mask Personally Identifiable Information (PII) across 1,000+ database columns using Fernet cryptography.
- Configured secure staging in Azure Blob Storage, schema management via Azure Cosmos DB (MongoDB API) and key management in Azure Key Vault.
- Built Azure Functions to provide event-driven, controlled unmasking of datasets for authorized consumer use cases.

Azure Data Engineer | Deloitte (Client: UBS AG)

London, UK | Jul 2023 - Aug 2023

- Developed data migration and archival pipelines moving legacy Credit Suisse applications and trade data into UBS Azure cloud infrastructure across 1,000+ core systems.
- Built Python-backed Azure Functions and Azure Data Factory (ADF) pipelines, accelerating ingestion throughput by 3x from PostgreSQL, Sybase and Oracle databases.
- Provisioned cloud resources using Terraform and deployed CI/CD workflows using GitLab CI/CD pipelines and Azure DevOps.
- Designed event-driven file validation, GPG encryption and archival workflows meeting global financial regulatory compliance and legal hold requirements.
- Built tracking dashboards using GitLab, Power BI and Alteryx to monitor migration status and pipeline health.

Senior Cloud Data Engineer | Deloitte (Client: Virgin Media O2)

London, UK | Jul 2022 - Apr 2023

- Architected cloud ingestion framework to load sensitive HR data for 40,000+ employee profiles from Oracle HCM, Workday and SuccessFactors into Google Cloud Platform.
- Enforced data security through GPG encryption, strict role-based access controls and custom Python modules for automated PII identification and masking.
- Developed shell scripts for automated GPG key generation and key rotation schedules, reducing key management overhead by 80%.
- Authored AbnInitio graphs and integrated dbt transformation workflows to standardize BigQuery data structures for downstream analytics.

Data Engineer | Deloitte (Client: Motor Insurers' Bureau)

London, UK | Mar 2021 - May 2021

- Engineered data quality scoring platform on Microsoft Azure for 3 core industry databases (MID, CUE and MIAFTR) covering 35M+ vehicle insurance records.
- Automated Azure environment deployment via Terraform, provisioning Azure Data Factory, Azure SQL Server, Storage Accounts and Informatica Secure Agents.
- Configured SFTP integration pipelines to ingest data feeds from Experian, CAP HPI and Solera into Azure SQL Server.
- Implemented Informatica Intelligent Cloud Services (IICS) data quality profiling and validation rules, generating automated scorecards for business stakeholders.

Senior Cloud & Data Engineer | HSBC

London, UK / Pune, India | Jan 2015 - Nov 2021

- GCP Migration (CESOP): Executed AWS to GCP migration proof-of-concept for CESOP financial payment reporting, writing Terraform scripts for GCP DataProc and Cloud Functions.
- Real-Time Streaming Platform: Built microservices self-service data platform; engineered near-real-time streaming processing 10M+ daily messages from Apache Kafka to BigQuery via GCS using Apache NiFi.
- Change Data Capture & Cross-Cloud Transfer: Implemented IBM CDC (DB2) and Attunity Replicate (Oracle) streaming via Kafka Connect and AWS Lambda for cross-cloud transfer between AWS and GCP.
- Legacy Data Migration & Reconciliation: Led source-to-target data transformation pipelines using IBM DataStage, Hadoop, Hive, Sqoop and Oracle for Common Data Repository (CDR) migration.

CERTIFICATIONS

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- **Google Cloud Certified: Professional Machine Learning Engineer** (Issued Dec 2025 - Dec 2027)
 - **Google Cloud Certified: Professional Data Engineer** (Issued Dec 2025 - Dec 2027)
 - **Google Cloud Certified: Generative AI Leader** (Issued Jan 2026 - Jan 2029)
 - **Google Cloud Certified: Associate Data Practitioner** (Issued Jan 2026 - Jan 2029)
 - **Google Cloud Certified: Cloud Digital Leader** (Issued Feb 2024 - Feb 2027)
 - **AWS Certified AI Practitioner** (Issued Sep 2024 - Sep 2027)
 - **HashiCorp Certified: Terraform Associate (003)** (Issued Sep 2024 - Sep 2026)

- **AWS Certified Cloud Practitioner** (Issued Jul 2022 - Feb 2028)
- **Microsoft Certified: Azure Fundamentals (AZ-900)** (2024)
- **Microsoft Certified: Data Fundamentals (DP-900)** (2024)
- **Oracle Cloud Infrastructure: Certified Foundations Associate** (2024)

EDUCATION & LANGUAGES

Bachelor of Engineering in Electronics and Communication | RGPV University, India (2010 - 2014)

Languages: English (Fluent), Hindi (Native)